

SHRAVANI MANEPALLI

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OBJECTIVE

Innovative and detail-oriented Senior Software Developer with a robust background in designing, building, and optimizing scalable backend systems and cloud-native applications. Adept at developing high-performance software solutions using Python, cloud platforms (AWS, Azure), RESTful APIs, and containerization tools like Docker. Experienced in leading end-to-end software development—from system architecture and coding to deployment and CI/CD integration. Passionate about solving complex technical problems, driving engineering excellence, and delivering impactful solutions that align with business goals.

EDUCATION

Texas Tech University	Masters of Science in Computer Science		2023-2025
Birla Institute of Technology And Science	Masters of Technology, Data Science	CGPA: 9.4	2020-2022
VNR Vignan Jyothi Institute of Engineering	Bachelors of Technology, Computer Science	CGPA: 9.3	2015-2019

SKILLS

GenAI technologies: LLMs, Model customization, LoRA Fine-tuning, Safety alignment, SFT, DPO, Stable diffusion, StyleGAN
Programming: Python, Flask, Django, C, C++, Java, SQL, MySQL, R, XML, JavaScript, React JS, Shell Scripting, JSON, Maven, Oracle
Deployment and Tools: AWS, Azure, Docker, Tableau, Qlik, InRule, Decisions, Unity, ML Studio , Azure Event Hub, AKS, Kubernetes
Engineering: Data Analysis, Design, ETL, Scaling, Computer Vision, Performance Testing, Distributed systems, Cloud Architecture
Backend and Data Science: SOAP and REST APIs, GenAI, Kafka, GIT, Jenkins, Elastic Search, Postman, Spark, Linux, Unix
Additional Skills: Agile, Scrum, Code Reviews , Team-oriented, Reliability,DesignPatterns,J2EE, Bash, JSP, Servlets,Security Measures

EXPERIENCE

TTUHSC — Senior Software Engineer 2024 — 2025

- Spearheaded automated invoice synchronization** between client SFTP servers and Finlogix’s processing system, improving invoice validation and classification.
- Built backend automation workflows using **Python, GPT-4o (LLM)**, and custom rule engines to summarize and classify invoice content, driving consistency and reducing manual errors.
- Developed RESTful microservices using Flask**, containerized with Docker, and deployed via Azure DevOps pipelines for rapid delivery and resilience.
- Enabled real-time validation and logging via **Azure EventHub and ElasticSearch-Kibana**, enhancing data visibility and auditability.
- Implemented schema validation, deduplication, and tabular integrity checks—leading to a 30% reduction in data processing time.

Valuelabs — CSC Service Works; Senior Software Engineer 2020 — 2023

- Designed and deployed end-to-end ML pipelines** for customer churn prediction and behavioral segmentation using Python and Scikit-learn, enabling data-driven retention strategies across business teams.
- Streamlined CI/CD workflows with Azure DevOps**, enhancing pipeline automation for faster and more reliable deployment of ML models into production.
- Built scalable data ingestion systems** leveraging Azure Event Hub, RESTful APIs, and ElasticSearch, reducing latency by 30% and enabling near real-time analytics and reporting.
- Modernized legacy infrastructure by migrating services to Azure Cloud, achieving higher availability, scalability, and operational efficiency.
- Integrated anomaly detection and robust validation logic into data pipelines, significantly improving data integrity and early fault detection in production workflows.
- Developed and optimized RESTful APIs** to support high-throughput data workflows, enabling seamless integration across microservices and faster business decision-making.

- **Developed backend applications in Python**, using PostgreSQL for structured data storage, efficient querying, and reporting.
- **Containerized services with Docker** and used Maven for streamlined builds, improving development consistency and deployment speed.
- Automated infrastructure provisioning on Azure using templates and scripts, enabling faster, reliable deployments.
- Conducted performance tuning and load testing on microservices in Linux environments, ensuring scalable, SLA-compliant systems.
- **Integrated Azure Event Hub** for real-time data ingestion, significantly improving visualization speed and responsiveness.
- **Configured Azure load balancers** and secure network tunnels to enhance cross-cloud communication, throughput, and reliability.

Intern — Research and Development Intern; Valuelabs

Jan 2019 — Mar 2019

- **Integrated InRule and Decisions rule engines** to build dynamic business logic for CSC ServiceWorks, enabling automated data categorization and advanced reporting.
- **Developed responsive frontend interfaces** using **HTML5, CSS3, and Bootstrap**, and deployed services on **Azure** for high availability and secure access.

ACADEMIC PROJECTS

Handwritten Recognition using Neural Networks

- Developed a high-accuracy CNN model for handwritten digit recognition using the **MNIST dataset**, achieving **>98% accuracy** and outperforming traditional machine learning classifiers.
- **Designed the architecture** using TensorFlow and Keras **with convolution, pooling, and dropout layers**.
- Preprocessed image data and used GPU acceleration to speed up training and improve convergence.
- **Validated model** using confusion matrix and accuracy metrics to ensure robustness.

Breast Cancer Detection using Machine Learning with SVM

- **Built a diagnostic ML pipeline** using **Support Vector Machines (SVM)** for binary classification of breast cancer from clinical datasets with **97.4% prediction accuracy**.
- Applied **PCA** and normalization for improved feature quality and model performance.
- Used **scikit-learn** for training, hyperparameter tuning, and **cross-validation** to avoid overfitting.
- Achieved strong results in **precision, recall, and F1-score**, outperforming baseline models.

LEADERSHIP EXPERIENCE AND ACTIVITIES

Awards in Valuelabs — Senior Software Engineer

- Awarded star of the quarter for Q1, Q3 2022-23 and Q3 2022 from Valuelabs for my contributions to the project.
- Ranked in top 10 for the Code of Good Hackathon conducted by JP Morgan Chase & Co.
- Organizing Member for Startup India League by Million Minds

Biomedical Outreach and Leadership Team — Member

- Planned and facilitated programs to guide youth towards higher education and stimulate interest in science and technology.
- Traveled 2 times per month to elementary schools in the local community to present 8 unique interactive experiments.

CERTIFICATIONS

- Google Data Analytics from Google
- Python certificate by Technosoft
- PL/SQL certificate by Oracle